

Optimization of Utility Consumption by Heat Integration and MTBE Column Analysis

MINI PROJECT 2
CHE 396 – SENIOR DESIGN I
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Project Charter

Project Title	Team Members
Optimization of Utility Consumption by Heat Integration and MTBE Column Analysis	Kathya Chavez Anna Borkowska Omar Takroui Esteban Vazquez-Elizondo
Problem 1 - Optimize Utility Consumption by Heat Integration	
Goals	Deliverables
- Analyze, understand, and accurately optimize utility consumption by heat integration - Illustrate the optimal process design for heat exchanger network	- Design basis and plan - PFD before and after heat integration - Composite curves - Calculation of $\Delta T_{\min}$ - Pinch Temperature calculation - Heat exchanger network design - Calculate utility costs before and after heat integration
In Scope	Out of Scope
- Design the optimized utility consumption by minimizing the costs - Reducing cost of utilities and using heat from other streams	- Optimization of existing processes - Safety protocol and environmental aspects
Problem 2 – MTBE Column Analysis	
Goals	Deliverables
- Analyze, understand, and accurately simulate in Aspen - Recovery of 99.5% of MTBE	- Design basis and plan - Block Flow Diagram for this Process - Feed and Mass Balance for Stages - PFD for Process - Aspen Simulation of MTBE Column
In Scope	Out of Scope
Design the appropriate equation of state	- Optimization of process - Heat Exchange Design - Safety protocol

Design Plan

Optimize Utility Consumption by Heat Integration and MTBE Column Analysis Design Plan					
Task ID	Task	Start Date	Duration (Days)	Finish Date	Assignee
Logistics		1-Oct	2	3-Oct	All
1	First Meeting (Task Delegation)	1-Oct	1	2-Oct	All
2	PFD Research	2-Oct	1	3-Oct	Esteban
3	Project Charter	2-Oct	1	3-Oct	Esteban
4	Design Plan	2-Oct	1	3-Oct	Esteban
5	Layout/Format	2-Oct	1	3-Oct	All
Problem 1 – Optimize Utility Consumption by Heat Integration Design		3-Oct	6	9-Oct	All
Introduction		3-Oct	2	5-Oct	Esteban
6	Objective	3-Oct	2	5-Oct	Esteban
Process Description		3-Oct	6	9-Oct	All
7	Problem	3-Oct	2	5-Oct	Esteban
8	Design Basis and Assumptions	3-Oct	3	6-Oct	Omar
9	PFD before and after heat integration	4-Oct	4	8-Oct	Esteban
10	Pinch Temperature Calculation, Optimizing $\Delta T_{\min}$ and Composite Curves	4-Oct	5	9-Oct	Anna
11	Heat Exchanger Network Design	4-Oct	5	9-Oct	Esteban, Kathya
12	Utility costs before and after heat integration	4-Oct	5	9-Oct	Anna, Kathya
Problem 2 – MTBE Column Analysis		3-Oct	6	9-Oct	Esteban, Kathya, Omar
Introduction		3-Oct	2	5-Oct	Esteban
13	Objective	3-Oct	2	5-Oct	Esteban
Process Description		3-Oct	6	9-Oct	Esteban, Kathya, Omar
14	Problem	3-Oct	2	5-Oct	Esteban
15	Design Basis and Assumptions	3-Oct	3	6-Oct	Esteban
16	Block Flow Diagram for this Process	4-Oct	4	8-Oct	Kathya
17	Feed and Mass Balance for Stages	4-Oct	5	9-Oct	Omar
18	PFD for Process	4-Oct	5	9-Oct	Esteban
19	Aspen Simulation of MTBE Column	4-Oct	5	9-Oct	Esteban
Wrap-Up		9-Oct	3	12-Oct	All
20	Section Unification	9-Oct	3	12-Oct	All
21	Proofreading / Plagiarism Check	10-Oct	3	12-Oct	All

Problem 1 – Optimize Utility Consumption by Heat Integration Design

Introduction

Objective

To optimize utility costs by integrating a heat exchanger network for a company. With the stream data given, we created the process flow diagram (See *Figure 1*) to get an idea of how to design the heat exchangers. Comparing the PFD before and after heat integration will give us a cost savings assessment to present to the company. After heat integration, composite curves can be created using the data of the optimized pinch temperature. Finally, utility costs can be calculated using the optimal utilities and processes to reduce costs.

Process Description

Problem

To develop cost savings to our customer by optimizing heat integration from a process that has no heat exchanger network. We researched the cost of the hot and cold utilities that can be used in our process and planned on designing a network that will reduce operational costs.

Design Basis and Assumptions

Our goal is to optimize the utility consumption of a process involving multiple heaters and coolers into a more energy efficient process by heat integration. The original process involves two feed streams mixing and passing through a heater before entering a reactor. The exit stream then passes through another heater and enters a distillation column equipped with coolers for the top and bottom streams. The energy released from the two coolers in the original process can be used to support heating the stream before the distillation column. A split stream is used following Heater 1 to account for the lower specific heat than the cooled streams. The original process requires 27,700 kW of energy input, whereas the optimized process only needs 6,100 kW. This more than 450% cut in energy consumption translates into massive cost savings totaling of a little under 2.1 billion dollars!

Our assumptions for this design are 85% efficiency heat exchangers, 100% efficiency motors (for heating and cooling), steady state fluid flow, specific heat independent of temperature, assuming 8000 hours of operation/year and constant temperatures.

PFD – Before and After Heat Integration

We were given a specific process and had to develop a simple model to know what the heat integration can be. From the figure on the next page (See *Figure 1*), we have 4 different utilities that will be needed. This includes 2 heaters for streams 3 and 5. Also, we need 2 coolers for streams 7 and 9.

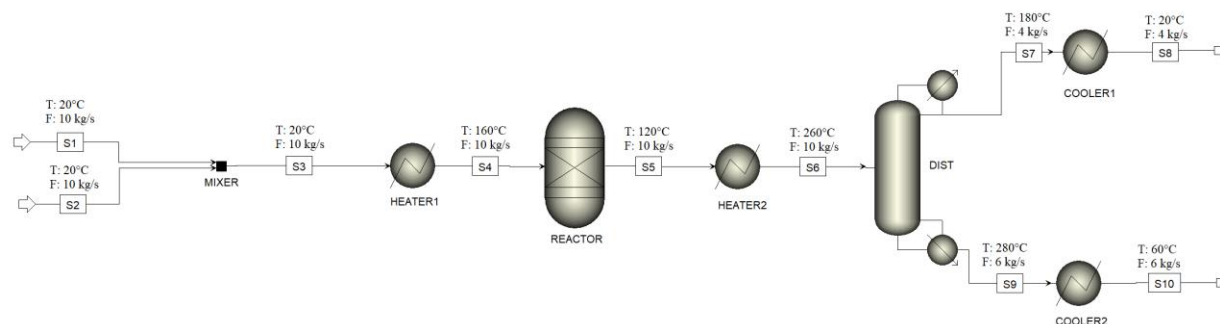


Figure 1 – PFD Before Heat Integration

With or without heat integration, the amount of heat needed is calculated with the pinch temperature equation:

$$Pinch\ Hot = T_{pinch} + \frac{\Delta T}{2} = 130 + \frac{20}{2} = 140\ ^\circ C$$

$$Pinch\ Cold = T_{pinch} - \frac{\Delta T}{2} = 130 - \frac{20}{2} = 120\ ^\circ C$$

To get the value of T_{pinch} , we can solve it using the Excel worksheet to figure out what the temperature is, and in this case, it came out to be 130 °C (See Figure 3). With the temperatures known, we can calculate above and below the pinch energies that are needed (See Table 1):

	Above Pinch	Below Pinch
	kW	kW
Stream 1	1600	4800
Stream 2	4200	2400
Stream 3	2000	5000
Stream 4	7700	-

Table 1 – Before Heat Integration

When we applied heat integration into the process, it will be using the heat energy of Cooler 1 and Cooler 2 to heat up Heater 2. The heat energy of Heater 1 will be split to cool down Cooler 1 and Cooler 2. We had to ensure that there was a split stream in Heater 1 as the CP value for the hot streams was lower than cold stream. The following process was designed in order to make the pinch analysis correct (See Figure 2):

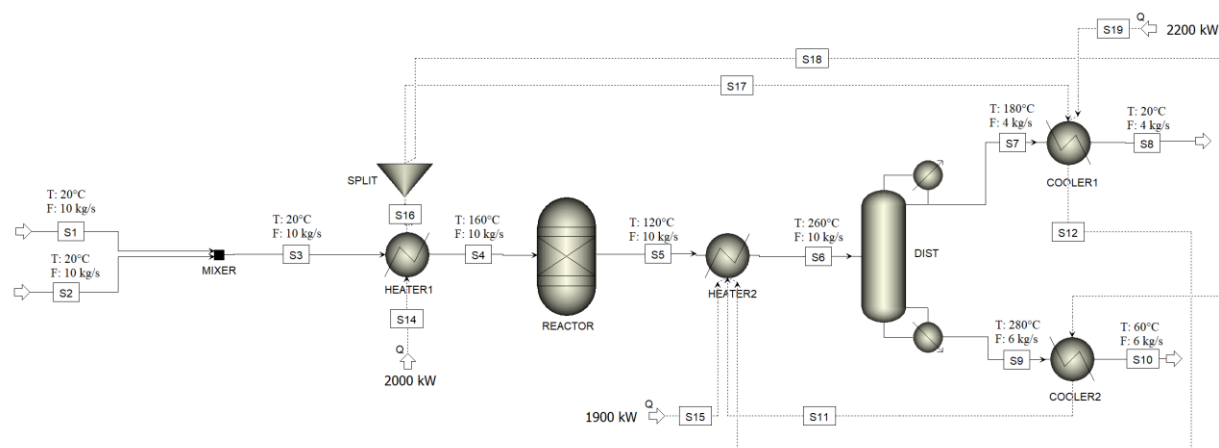


Figure 2 – PFD After Heat Integration

After heat integration, the total input energy is only 6,100 kW in comparison to before integration of 27,700 kW – almost a whopping 450% cut! This will yield higher cost savings to the company which will be explained in more detail in the Utility Cost section. Although there are other designs that can be used for the heat exchange network, we decided to go this route as the parameters of utility were similar. We need 2000 kW, 1900 kW, and 2200 kW of energy which are all an average of about 2000 kW. This means we can work with similar pipes in the facility and can reduce capital costs of installing new equipment or pipelines that can handle the transfer. If we have different types of energies throughout the plant, this can mean more costs on maintenance and ensuring we have the correct equipment to handle this.

Pinch Temperature, Optimization, and Composite Curve

The pinch temperature is an essential component in the heat exchanger network design because it divides the system into two thermodynamic regions.^[1] The area above the pinch temperature is considered to be the heat sink where heat flows in from the hot utility but no heat flows out.^[1] The opposite is true for the area below the pinch, heat flows out of the region to the cold utility and it acts as a heat source.^[1] There is no heat flowing across the pinch because the two regions are energetically balanced. Therefore, using the pinch temperature allows the process to be decomposed into two sub problems.^[1]

In order to calculate the pinch temperature, a minimum change in temperature is given as 20 °C. Although the minimum change in temperature can be calculated using composite curves, in this problem it is given. Using the Problem Table Excel spreadsheet^[2] the four process-stream source-and-target temperatures and heat capacity flow rates were inputted. The change in minimum temperature was also inputted and then the Excel spreadsheet calculates the pinch temperature. The pinch temperature is taken from where kW is zero in the Problem Table section of Figure 3. It can be seen in the highlighted section that the Pinch Temperature is 130 °C.

1. Minimum temperature approach

$\Delta T_{\min}$ 20 °C

2. Stream data

Stream No.	Actual temperature (°C)		Interval temperature (°C)		Heat capacity flow rate CP (kW/°C)	Heat load (kW)
	Source	Target	Source	Target		
1	20	160	30	170	50	7000
2	120	260	130	270	55	7700
3	180	20	170	10	40	6400
4	280	60	270	50	30	6600
5	0	0	10	10	0	
6	0	0	10	10	0	
7	0	0	10	10	0	
8	0	0	10	10	0	

3. Problem table

Interval	Interval temp (°C)	Interval ΔT (°C)	Sum CP_c - sum CP_h (kW/°C)	dH (kW)	Cascade (kW)	(kW)
	270				0	3900
1	270	0	0	0	0	3900
2	170	100	25	2500	-2500	1400
3	170	0	25	0	-2500	1400
4	130	40	35	1400	-3900	0
5	50	80	-20	-1600	-2300	1600
6	30	20	10	200	-2500	1400
7	10	20	-40	-800	-1700	2200
8						

Figure 3 – Pinch Temperature

As mentioned above, the change in minimum temperature was given for this problem. However, the following will explain how to calculate the change in minimum temperature using composite curves. The first thing that was done was to plot temperature vs. enthalpy for the two hot streams in the system (see Figure 4). Then the two hot streams are combined by taking out the points where the two streams are parallel to each other and replacing that with a trend line that connects the ends of the two streams (see Figure 5). The same procedure is then repeated for the two cold streams (see Figure 6 and 7). The combined hot and cold streams are then combined onto the same composite curve (see Figure 8). It can be seen in Figure 8 where the hot, cold, and required utilities are. It can also be seen that the change in minimum temperature is the point where the hot stream line indents towards the cold stream point. From this composite curve, the change in minimum temperature is about 10 °C. The optimized minimum change in temperature can be calculated by moving the two lines horizontally in and away from each other and either increasing or decreasing the change in temperature.^[2] By moving the lines horizontally, the hot and cold utilities are changed. From there, the utilities at different minimum change in temperatures can be compared and it can be determined what the optimized minimum change in temperature is 40 °C for hot stream and about 30 °C for cold stream .^[2]

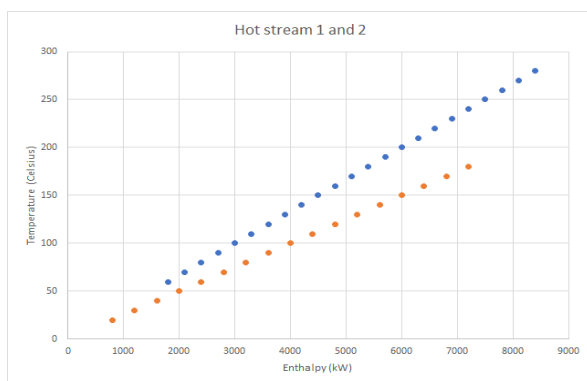


Figure 4 – Hot Stream 1 and 2

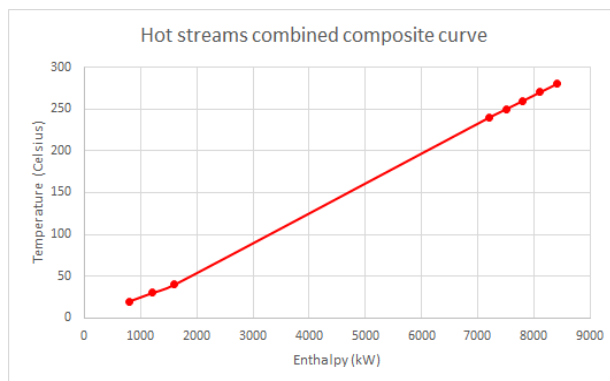


Figure 5 – Hot Stream Composite Curve

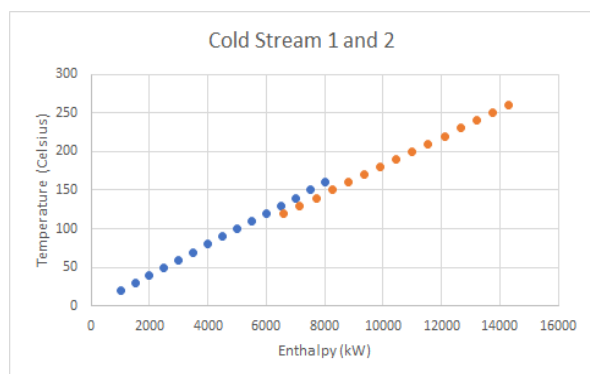


Figure 6 – Cold Stream 1 and 2

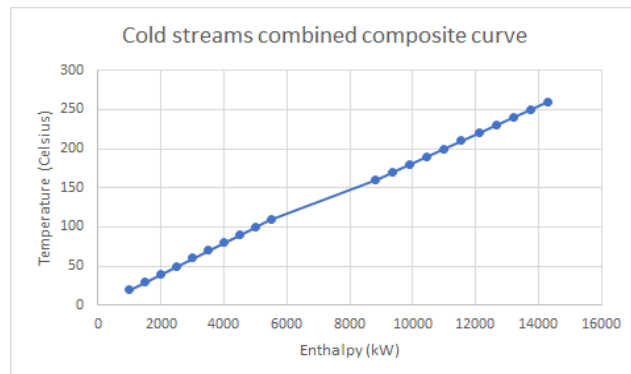


Figure 7 – Cold Stream Composite Curve

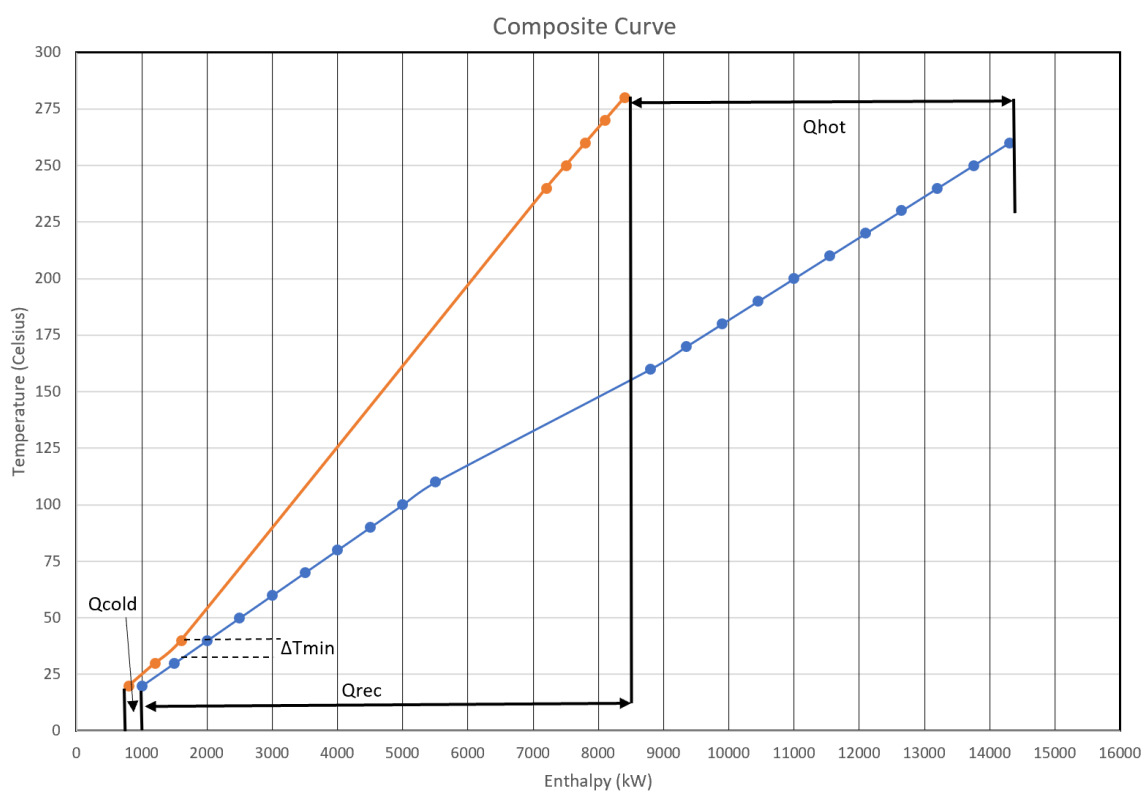


Figure 8 – Composite Curve

Heat Exchanger Network Design

Heat exchanger network creates an optimization of heat transfer in a system. Combining streams above pinch point (140 °C) could change temperature with heaters while cold streams have cold temperatures to hot temperature. A cool heat exchanger is also added to hot stream to share temperature while also sharing heat when streams are going hot to cold streams. Below is our design for heat exchange network (See Figure 9):

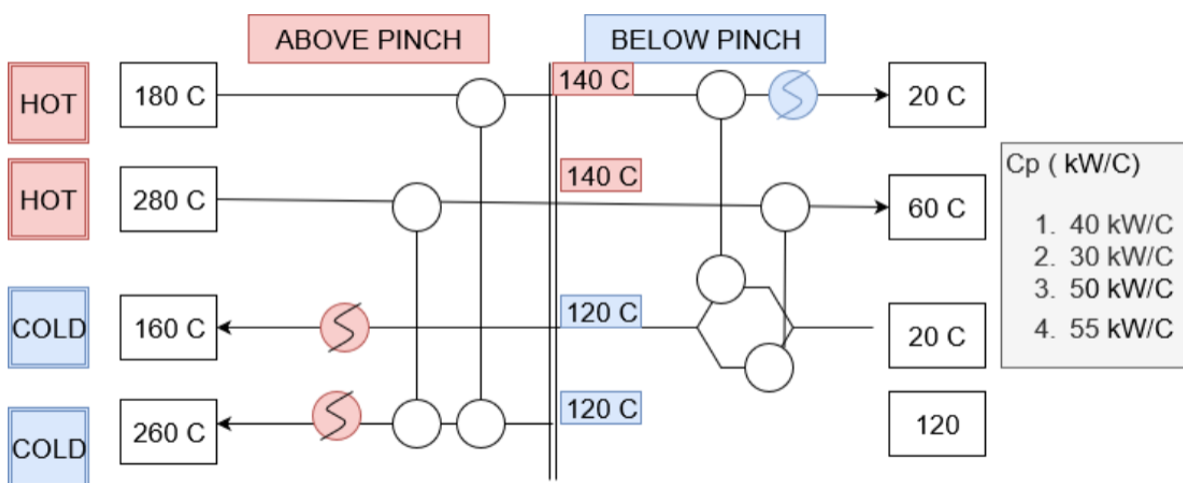


Figure 9 – Heat Exchanger Network with Pinch Analysis

We need to calculate the change in enthalpy so from thermodynamics:

$$Q = C_h(T_{ti} - T_{h0})$$

$$Q = C_c(T_{c0} - T_{ci})$$

$$\text{Solve for } \frac{Q}{C_h} \text{ and } \frac{Q}{C_c}$$

$$(T_{ti} - T_{h0}) = \Delta T_2$$

$$(T_{c0} - T_{ci}) = \Delta T_1$$

$$\Delta T_{min} = \Delta T_1$$

$$\Delta T_2 = \frac{Q(C_c - C_h)}{C_c * C_h} + \Delta T_{min}$$

$$\Delta T_2 \geq \Delta T_{min}$$

$$\text{Pinch: } \Delta T_2 - \Delta T_1 = \frac{Q(C_c - C_h)}{C_c * C_h}$$

$$\Delta H = M * Cp * \Delta T$$

Above pinch

$$STREAM\ 1 = 40 (180 - 140) = 1600\ kW$$

$$STREAM\ 2 = 30 (280 - 140) = 4200\ kW$$

$$STREAM\ 3 = 50 (160 - 120) = 2000\ kW$$

$$STREAM\ 4 = 55 (260 - 120) = 7700\ kW$$

Below Pinch

$$STREAM\ 1 = 4800\ kW$$

$$STREAM\ 2 = 2400\ kW$$

$$STREAM\ 3 = 5000\ kW$$

$$STREAM\ 4 = 0\ kW\ (\text{since temp. is at } 120^{\circ}\text{C})$$

Split Stream

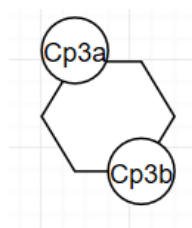


Figure 10 – Zoomed in Cold Stream Split – Showing Cp3a and Cp3b

$$Cp_{3a} = \frac{2600\ kW}{100\ \Delta T} = 26\ \frac{kW}{^{\circ}\text{C}}$$

$$Cp_{3b} = \frac{2400\ kW}{100\ \Delta T} = 24\ \frac{kW}{^{\circ}\text{C}}$$

Hot stream 1: CP value is 40 kW/°C and cold stream 3: CP value is 50 kW/°C. Since below the pinch, $CP_{Hot} \geq CP_{Cold}$ we need to split the third stream to satisfy this requirement as they are not in agreement now. The total power needed in stream 3 is 5000 kW, we can take stream 2 and transfer the energy to stream 3. This will leave stream 3 with 2600 kW needed. We can complete this power by transferring energy from stream 1. This leaves our design with a cooler needed to stream 1 of 2200 kW. But once we calculate the new CP_{Cold} values, CP_{3a} is 26 kW/°C and CP_{3b} is 24 kW/°C which are now less than the hot CP value of 40 kW/°C which satisfies the requirement.

$$CP_{3a} + CP_{3b} = CP_{3(Total)}$$

$$CP_{Hot} \geq CP_{Cold(3a)} = 40\ \frac{kW}{^{\circ}\text{C}} \geq 26\ \frac{kW}{^{\circ}\text{C}}$$

$$CP_{Hot} \geq CP_{Cold(3b)} = 40\ \frac{kW}{^{\circ}\text{C}} \geq 24\ \frac{kW}{^{\circ}\text{C}}$$

Utility Cost – Before and After Heat Integration

Before		Power						
		kW		kJ/yr	mj/yr			cost
180-20	Stream 1	6400		1.8432E+11	184320		\$ 4.50	\$ 829,440.00
				kJ/yr	gal water /1000			
280-60	stream 2	6600		205714285.7	54278.17565		\$ 0.10	\$ 5,427.82
				kJ/yr	KG steam/yr	Lb/1000	Cost	
160-20	Stream 3	7000		2.016E+11	106723133.9	235284.0623	\$ 5.47	\$ 1,287,003.82
			eff.	mmBtu/h	mmBtu/year		Cost	
260-120	Stream 4	7700	9058.824	30.91010178	247280.8143		\$ 3.20	\$ 791,298.61
							sum total	\$2,913,170.24
After								
				kJ/yr	mj/yr		cost per 1000	
cooler	cooler	2000		57600000000	57600		\$ 4.50	\$ 259,200.00
				kJ/yr	KG steam/yr	Lb/1000	Cost	
above	heater	2200		63360000000	33541556.38	73946.41957	\$ 5.47	\$ 404,486.92
			eff.	mmBtu/h	mmBtu/year		Cost	
below	heater	1900	2235.294	7.627167972	61017.34378		\$ 3.20	\$ 195,255.50
							sum total	\$ 858,942.42
							saving	\$2,054,227.83

Figure 11 – Heat Exchanger Cost Comparison

With simple conversion, changing power that assumed the system is running 8000 hr./year. Considering that every system steam has either steam, water, or vapor dependent on pressure. The cost of a heated stream 180 °C to 20 °C, reflected on the utility cost calculation stream 1, 6400kW at a cost of \$4.50 MJ/year, the sum would be \$829,440.00.^[1] Power of 6000kW, hot stream of water the cost per 1000 gallons is at a cost of \$0.10, total cost is \$5,427.82. Cold stream, 20 °C to 160 °C, at \$5.47, cost \$1,287,003.82 per year. Finally, the power efficiency of 85% of 7700kW, the cost is \$3.20 converts to \$791,298.61.

After heat exchange analysis, cooler applied to hot stream 180 °C to 20 °C costs \$259,200.00, which already has a difference \$570,240.00 cost of savings. The first heater on the cold stream 20 °C to 160 °C cost is \$5.47 thus total expense is \$404,486.92, which saves \$882,516.91 from before heat exchanger analysis. The second heater cost of expense is \$195,255.50, the saving difference is \$596,043.11. With total costs considered for heat exchange analysis costs \$858,942.42, total difference is \$2,054,227.83 in savings (See Figure 1!).

Problem 2 – MTBE Column Analysis

Introduction

Objective

Methyl tertiary butyl ether (MTBE) is extracted from unreacted methanol and C₄ compounds by distillation. However, in the reactor that we are simulating in Aspen, we are to get a recovery percentage of 99.5% MTBE for the product. We were to play around with the number of stages, tray positioning, reflux ratio, tray positioning and other parameters to see which one more cost will be effective and get us the percentage needed.

Process Description

Problem

There are many equations of state that can be used but we need to ensure it will be compatible with the process we are using of methanol conversion to MTBE. We are also to change parameters and plot the concentration profiles of MTBE product.

Design Basis and Assumptions

For the design basis, we used Peng-Robinson as our equation of state in the process. Reason for this was that other equations of state did not meet the required product purity to be 99.5% MTBE recovery. Since we are dealing with hydrogen bonding, we would think Wilson or NRTL will be correct but as we increased the pressure, there was little change in meeting the required purity of MTBE along with production rate of 7000 kg/h. For this reason, we went with Peng-Robinson.

Block Flow Diagram for this Process

Below is a simple block flow diagram of production of MTBE from isobutene and methanol (Figure 12):

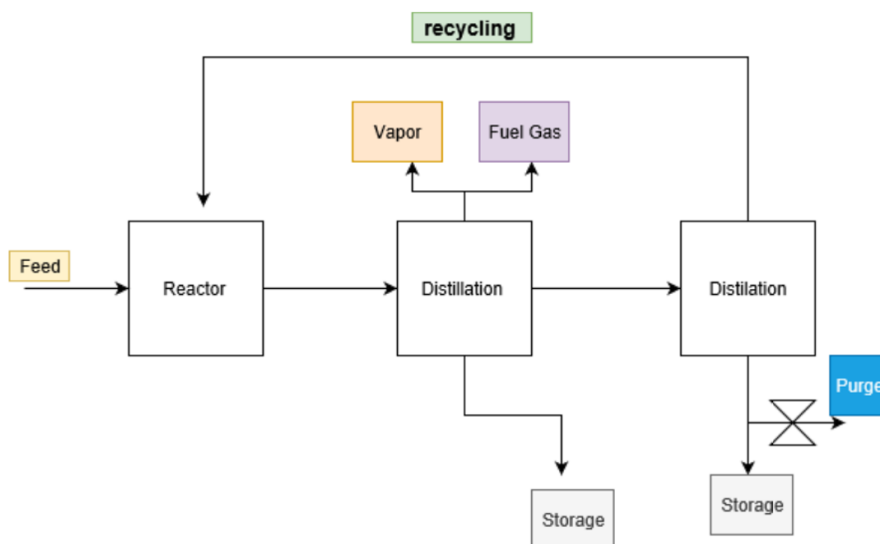
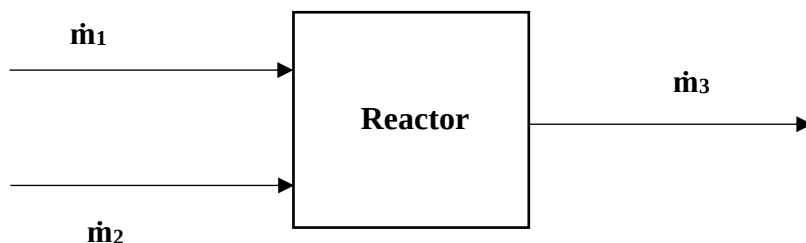


Figure 12 – Block Flow Diagram

Feed and Mass Balance for Stages



Reactor Material Balance:

Methanol:

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

Isobutene:

$$X_{A,3} \dot{m}_3 = 0.10 \dot{m}_1$$

C₄ Compounds:

$$X_{B,2} = 0.49$$

MTBE:

$$X_{C,2} = 0.51$$

$$X_{D,3} = 0.97(X_{B,2} \dot{m}_2) = 0.97(0.49 \dot{m}_2) = 0.475 \dot{m}_2$$

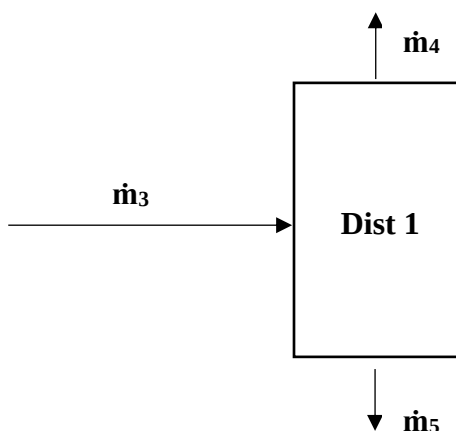
Let A = Methanol

B = Isobutene

C = C₄ Compounds

D = MTBE

E = Water



Distillation 1 Material Balance:

Methanol:

$$\dot{m}_3 = \dot{m}_4 + \dot{m}_5$$

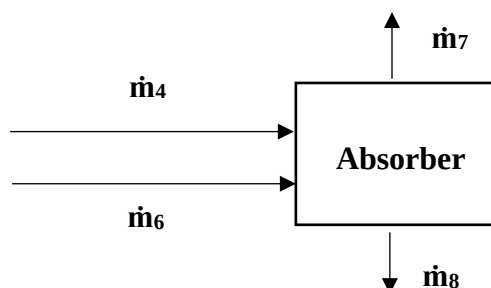
C₄ Compounds:

$$X_{A,4} \dot{m}_4 = 0.10 \dot{m}_1$$

MTBE:

$$X_{C,4} \dot{m}_4 = 0.51 \dot{m}_2$$

$$\dot{m}_5 = 0.995 X_{D,3} \dot{m}_3 = 0.995(0.475 \dot{m}_3) = 0.473 \dot{m}_3$$



Absorber Material Balance:

$$\dot{m}_4 + \dot{m}_6 = \dot{m}_7 + \dot{m}_8$$

Methanol:

$$X_{A,4} \dot{m}_4 = X_{A,8} \dot{m}_8$$

C₄ Compounds:

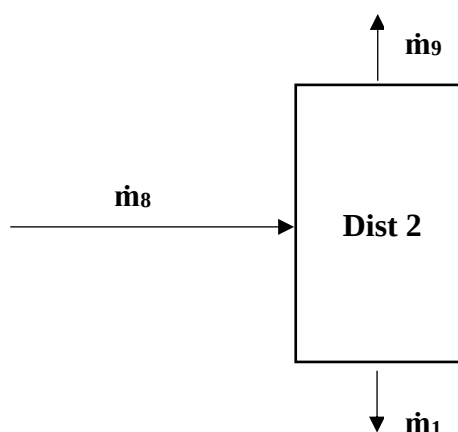
$$X_{C,4} \dot{m}_4 = \dot{m}_7$$

MTBE:

$$X_{D,4} \dot{m}_4 = X_{D,8} \dot{m}_8$$

Water:

$$\dot{m}_6 = X_{E,8} \dot{m}_8$$



Distillation 2 Material Balance:

$$\dot{m}_8 = \dot{m}_9 + \dot{m}_{10}$$

Methanol:

$$X_{A,8} \dot{m}_8 = \dot{m}_9$$

Water:

$$\dot{m}_{10} = X_{E,8} \dot{m}_8$$

MTBE:

$$X_{D,8} \dot{m}_8 = X_{D,10} \dot{m}_{10}$$

PFD for Process

Below is the PFD for the process that shows the shortcut model (top-half of *Figure 13*) and the rigorous model (bottom-half of *Figure 13*):

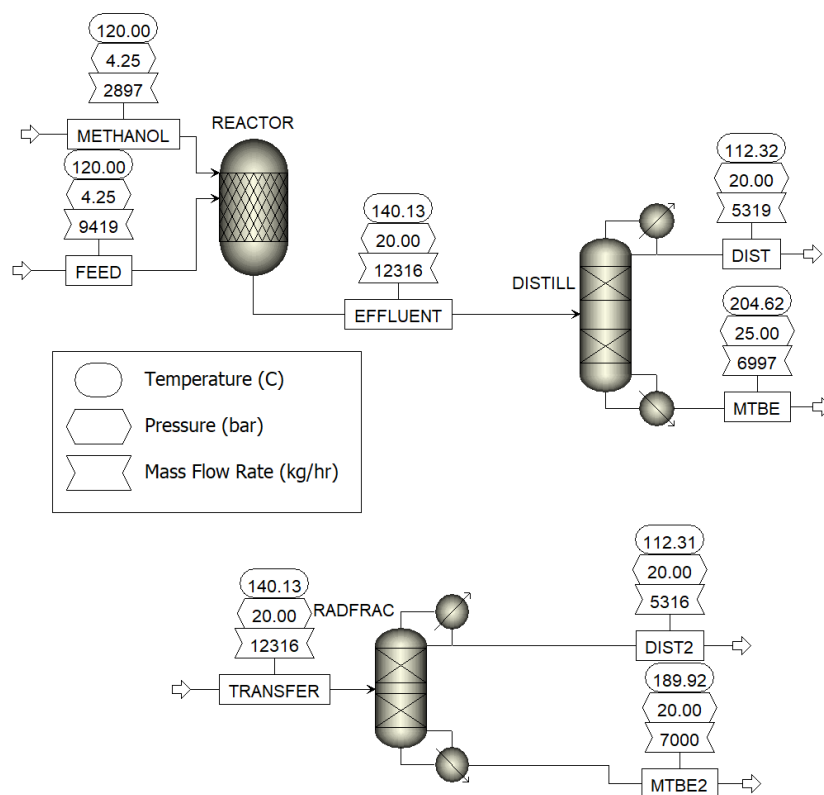


Figure 13 – PFD for Process

Aspen Simulation of MTBE Column

WILSON and NRTL equations of state were used in the Aspen simulation. However, they were unable to give us values of the target range of 99.5% MTBE recovery in the product stream. The PENG-ROB equation of state was the model that gave us a recovery that was greater than 97% in comparison with the other EOS. Under the Peng-Robinson EOS, we were able to get the binary analysis of both components with a pressure of 20 bar (See *Figure 14*):

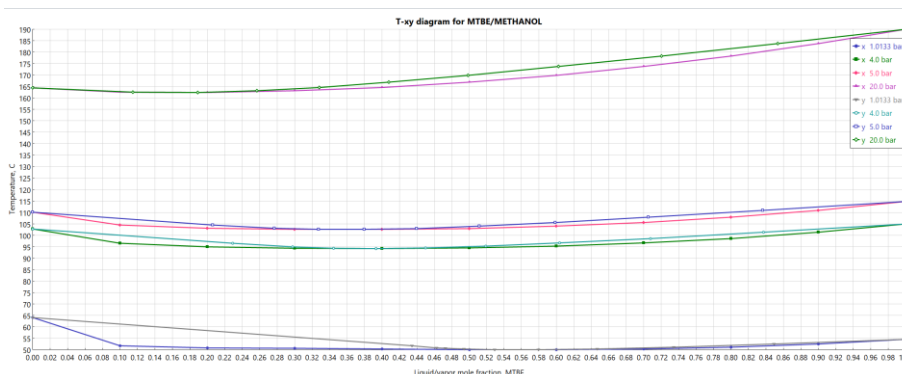


Figure 14 – Temperature and Mole Fraction of MTBE/Methanol

The mole fraction of MTBE/Methanol at .99 for 20 bar gives us a value of temperature of approximately 189 °C which is in correlation to our rigorous model operating at 20 bar (See *Figure 13 – bottom half*). This can conclude that the Peng-Robinson model will give us correct values and the purity we are looking for.

Azeotrope research for the distillation under Peng-Robinson gave us:

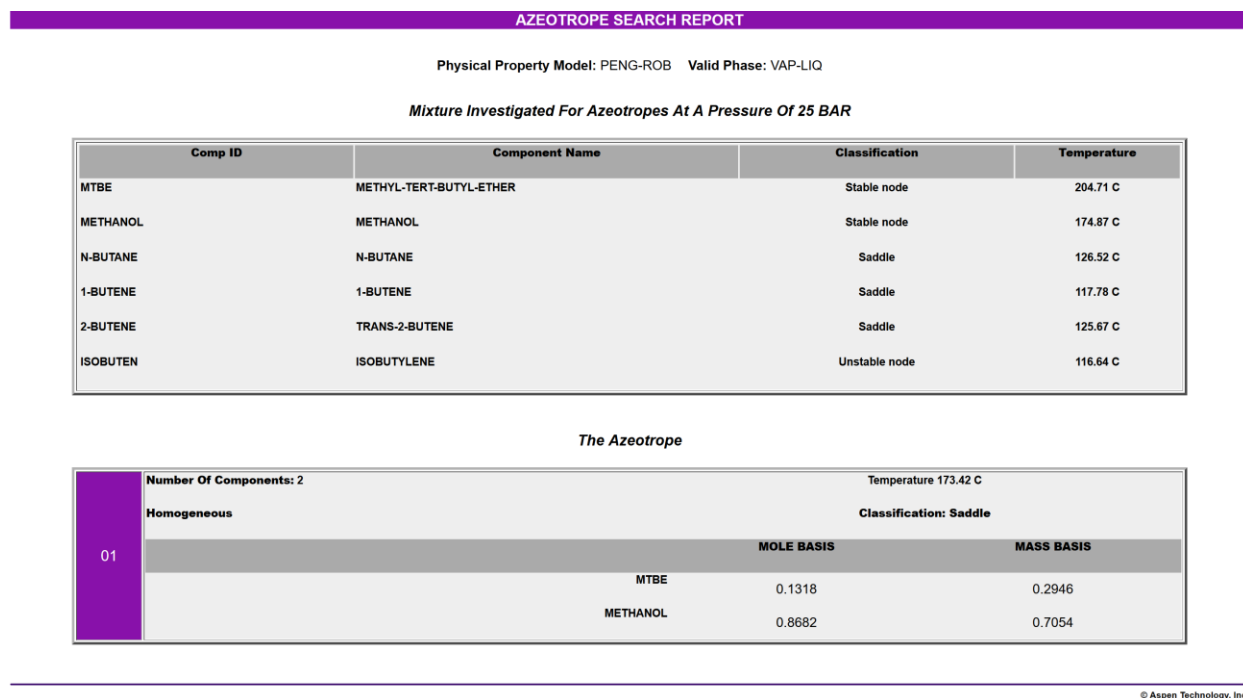


Figure 15 – Mixture Investigation for Azeotropes at 25 Bar

The above figure shows us that operating at higher pressure of 25 bar will push the azeotropic temperature higher and give us a better separation of the MTBE/Methanol. Once we reach the temperature of 173.42 °C, we will be stuck with that activity coefficient of the two components but as we have learned from the product stream from our simulation, we were able to get 99.86% mole fraction of MTBE for the rigorous model. This means that the mole basis shown in the azeotropic model in to top figure, does not correlate with the mole composition of product.^[3] When we are adding C₄ components in the distillation, MTBE is separated due to having a low activity coefficient of 0.1318 in comparison to methanol of 0.8682. This means that the distillation will be pure under a high-pressure vessel to push the azeotrope.^[4] On the design basis of distillation, other research groups have used Peng-Robinson equation of state and NRTL but when trying to get high purity above 97%, the pressure change did not affect the purity of MTBE under NRTL.^[5] The use of Peng-Robinson will give higher purity, but we are still dealing with high pressure distillation columns that will need to be depressurized for the storage or transport of product.

We ran the simulation with the shortcut model which gave us values for the product MTBE recovery along with other helpful parameters such as pressure, reflux ratio, distillate to feed ratio, number of stages, and feed stream stage. The shortcut model is shown below (See *Figure 16*):

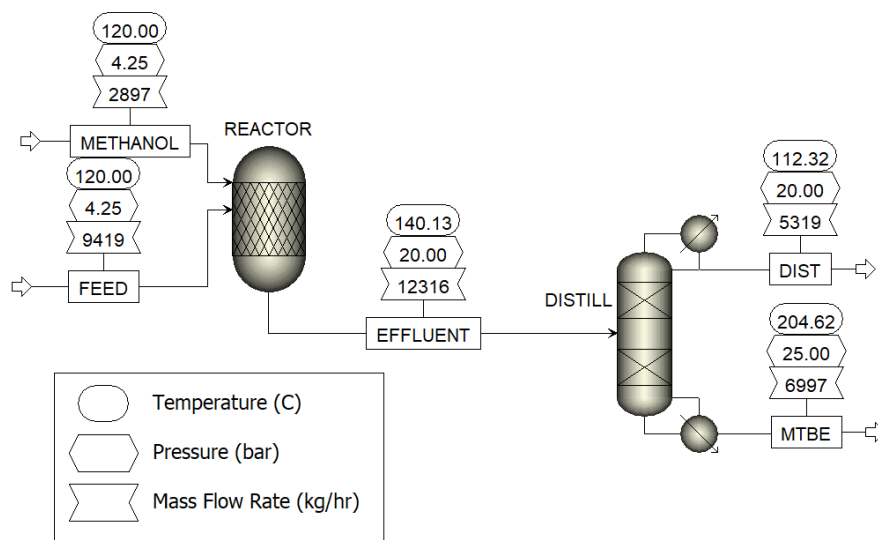


Figure 16 – Shortcut Model of MTBE Column

The distillation gave us the following results in the column:

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
						MTBE
Phase						Liquid
Temperature						204.618
Pressure						25
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.45939
Molar Density						0.0049097
Mass Density						0.432413
Average MW						88.0731
+ Mole Flows						79.4438
- Mole Fractions						
MTBE						0.998621
METHANOL						0.00134508
N-BUTANE						2.87285e-06
1-BUTENE						1.00319e-05
2-BUTENE						2.10544e-05
ISOBUTEN						3.77245e-07
+ Mass Flows						6996.87

Summary	Balance	Reflux Ratio Profile	Status
Minimum reflux ratio	1.96603		
Actual reflux ratio	2.26093		
Minimum number of stages	27.8657		
Number of actual stages	58.6602		
Feed stage	30.8864		
Number of actual stages above feed	29.8864		
Reboiler heating required	380824	cal/sec	
Condenser cooling required	318476	cal/sec	
Distillate temperature	112.321	C	
Bottom temperature	204.618	C	
Distillate to feed fraction	0.554784		

Figure 17 – Shortcut Model Results

We can use this data in order to model the rigorous model of the column design and see if we can make the concentration of product MTBE as close to 100% purity. The shortcut model gives us the ability to call out the percent of product we want out and will iterate to give us values of the amount of stages, above the feed stage, temperatures, and reflux ratio. Using this model, we can have an idea of what operational inputs we will need for the rigorous model (See *Figure 18*).

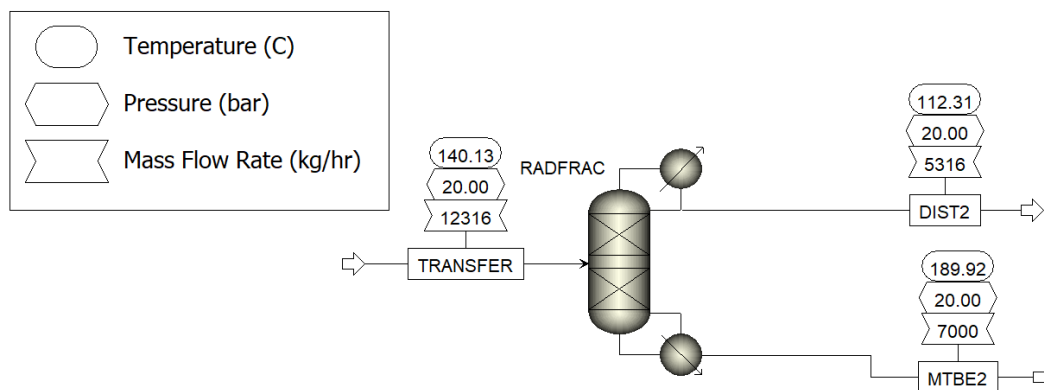


Figure 18 – Rigorous Model of MTBE Column

The rigorous model was easy to transfer the information from the shortcut model as we had an idea of what parameters we need in order to produce the target 7000 kg/hr mass flow rate of product and close to 100% purity of MTBE (See Figure 19).

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
						MTBE2
Phase						Liquid
Temperature			C			189.917
Pressure			bar			20
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy			cal/gm-K			-1.48583
Molar Density			mol/cc			0.00537205
Mass Density			gm/cc			0.473345
Average MW						88.1126
+ Mole Flows			kmol/hr			79.4438
- Mole Fractions						
MTBE						0.999336
METHANOL						0.000657808
N-BUTANE						6.31941e-07
1-BUTENE						1.53587e-06
2-BUTENE						4.18727e-06
ISOBUTEN						5.43159e-08
+ Mass Flows			kg/hr			7000

Figure 19 – Rigorous Model Distillation Results

The composition model of the rigorous model is in the next page which indicates clear separation at stage 30 (See Figure 20). The mole fraction of MTBE goes up as we have more stages and will be dependent on customer specifications. In our case, we are trying to get pure MTBE so this profile will satisfy that requirement. The temperature profile also shows that at stage 30, we have a rapid increase in temperature and the methanol and MTBE is separated (See Figure 21).

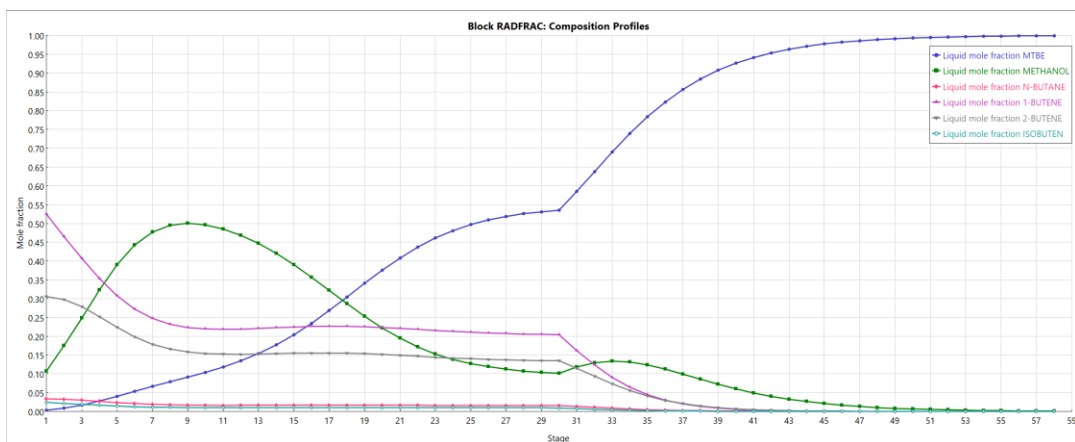


Figure 20 – Rigorous Model Composition Profile

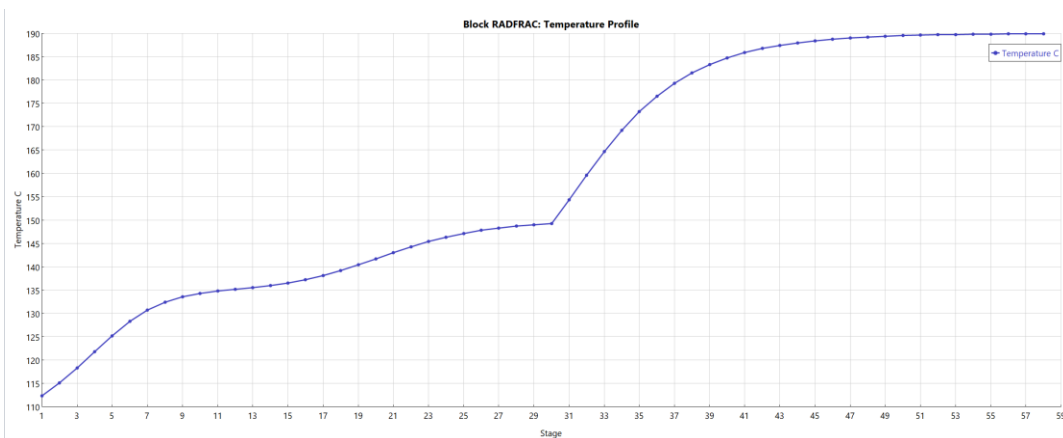


Figure 21 – Rigorous Model Temperature Profile

When we ran the model the mass flow rate was the target of 7000 kg/hr and our mole fraction of MTBE is 99.93% which is as close as we can get with the pressure of 20 bar. This meant that the number of stages we chose for our model was accurate (See Figure 22).

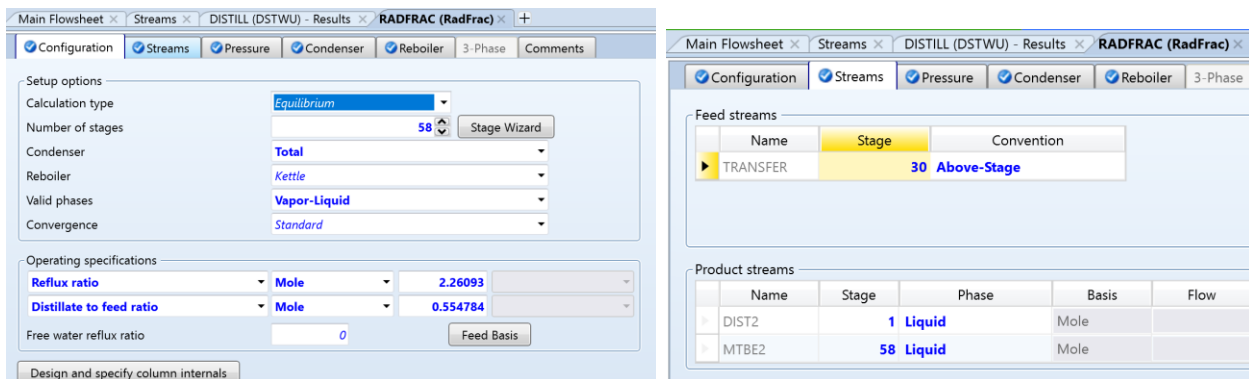


Figure 22 – Rigorous Model Parameters

Now, we need to investigate what the causes and effects are by changing the parameters such as reflux ratio, distillate to feed ratio, feed stream location and give us different concentration profiles. From these profiles, we can advise the effect of raising or lowering those parameters and consult with any company that has a similar model on the amount of stages or any other relation to the column creation to reduce costs.

Decreasing and Increasing Number of Stages (from 58 stages)

Decreasing Stages to 45:

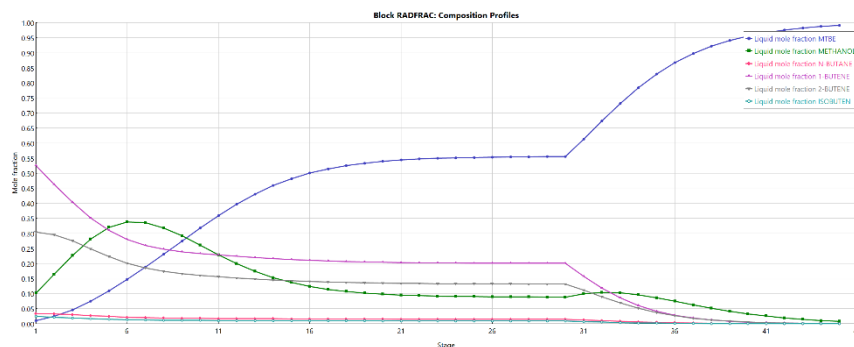


Figure 23 – Composition Profile and Product Results (45 Stages)

Increasing Stages to 65:

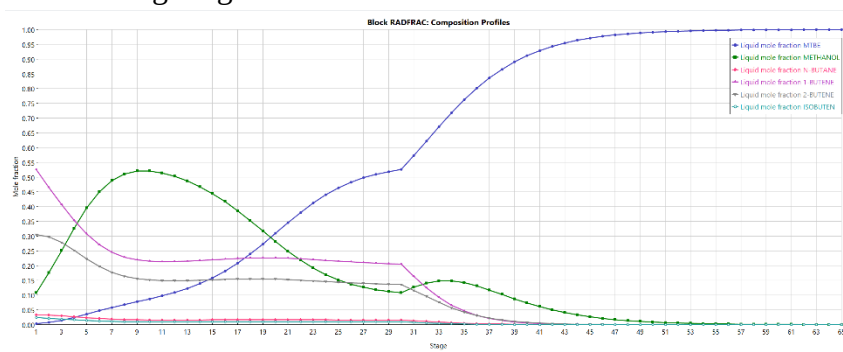


Figure 24 – Composition Profile and Product Results (65 Stages)

When we decrease the number of stages to 45 as in Figure 23, we get slightly lower temperature of 189.4 °C and the mole fraction of MTBE is 99.15%. In Figure 24, an increase in the number of stages to 65 the temperature is slightly higher to 190 °C and the mole fraction of MTBE is 99.98%. We can conclude that an increase of stages in the distillation column, we will get a higher purity of MTBE. The trade-off will be that using more stages will result in a higher footprint of the distillation column in the facility. This will also mean the capital cost will be higher. Depending on the customer demand, we need to find the customer specifications in order to provide the ideal number of stages.

Decreasing and Increasing Reflux Ratio (from 2.26093)

Decreasing RR to 1.5:

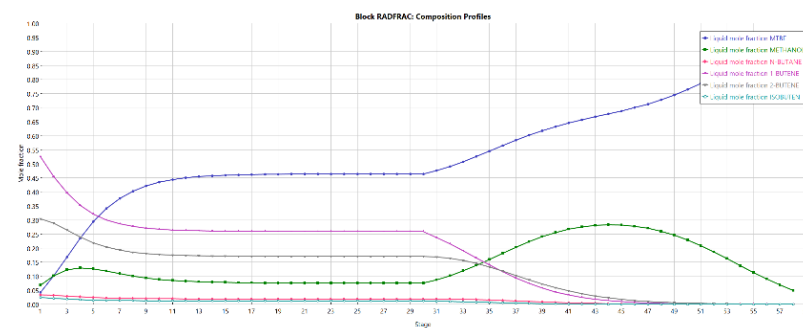


Figure 25 – Composition Profile and Product Results (RR 1.5)

Material	Vol% Curves	Wt% Curves	Petroleum	Polymers	Solids	Status
						MTBE2
Phase						Liquid
Temperature						186.774
Pressure						20
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.48358
Molar Density						0.00565578
Mass Density						0.482798
Average MW						85.3637
+ Mole Flows						82.0021
- Mole Fractions						
MTBE						0.950207
METHANOL						0.0494721
N-BUTANE						3.00956e-05
1-BUTENE						6.14798e-05
2-BUTENE						0.00022764
ISOBUTENE						2.08416e-06
+ Mass Flows						7000

Increasing RR to 3.5:

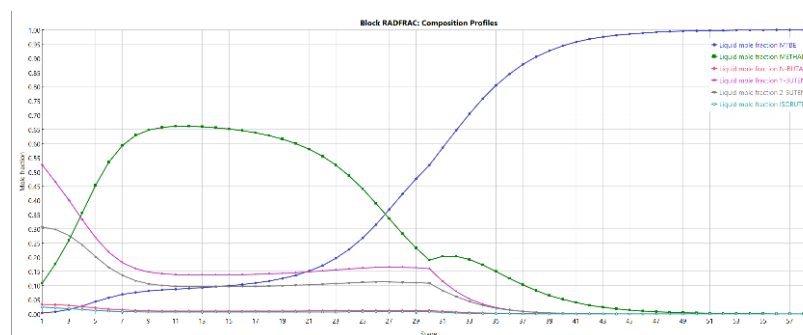


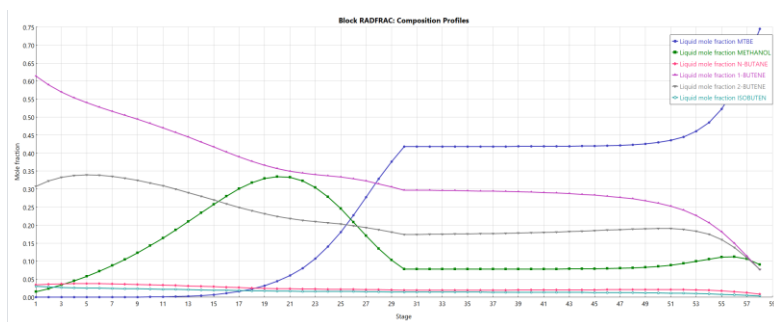
Figure 26 – Composition Profile and Product Results (RR 3.5)

Material	Vol% Curves	Wt% Curves	Petroleum	Polymers	Solids	Status
						MTBE2
Phase						Liquid
Temperature						189.942
Pressure						20
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.48588
Molar Density						0.00536983
Mass Density						0.473267
Average MW						88.1345
+ Mole Flows						79.424
- Mole Fractions						
MTBE						0.999729
METHANOL						0.000269672
N-BUTANE						1.06563e-07
1-BUTENE						2.56414e-07
2-BUTENE						7.11509e-07
ISOBUTENE						9.02553e-09
+ Mass Flows						7000

Decreasing the reflux ratio (RR) to 1.5, the purity of MTBE is 95.02% with a lower temperature of 186.7 °C as shown in Figure 25. If we increase the RR to 3.5, the purity of MTBE is 99.97% with a temperature of 189.9 °C as shown in Figure 26. We can conclude from this that as we increase the reflux ratio, we will be getting a higher purity of MTBE production which is ideal. Essentially, we are decreasing the number of stages the higher the reflux ratio is. This will mean we can develop a trade-off between the number of stages and temperature needed to provide purity. Higher temperatures will mean specialty equipment that can endure the high temperature and pressures.

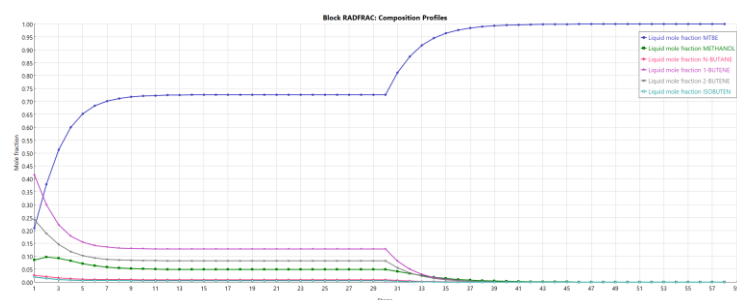
Decreasing and Increasing Distillate to Feed Ratio (from 0.554783528)

Decreasing DF to 0.4:



Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
MTBE2						
Phase						Liquid
Temperature						167.967
Pressure						20
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.44624
Molar Density						0.00641582
Mass Density						0.499291
Average MW						77.8219
♦ Mole Flows						89.949
— Mole Fractions						
MTBE						0.744726
METHANOL						0.0899973
N-BUTANE						0.00872367
1-BUTENE						0.0763806
2-BUTENE						0.0769549
ISOBUTENE						0.00321724
♦ Mass Flows						7000

Increasing DF to 0.7: *Figure 27 – Composition Profile and Product Results (DF 0.4)*



Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
MTBE2						
Phase						Liquid
Temperature						189.96
Pressure						20
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.48592
Molar Density						0.00536834
Mass Density						0.473215
Average MW						86.1493
♦ Mole Flows						79.4108
— Mole Fractions						
MTBE						0.999992
METHANOL						7.46677e-06
N-BUTANE						1.68983e-08
1-BUTENE						4.41392e-08
2-BUTENE						1.07532e-07
ISOBUTENE						1.58288e-09
♦ Mass Flows						7000

Figure 28 – Composition Profile and Product Results (DF 0.7)

Decreasing the distillate to feed ratio (DF) will result in low purity of product stream. In *Figure 27*, DF is set to 0.4 and we get a purity value 74.5% MTBE with a low temperature of 168 °C. Now if we increase DF to 0.7, we get no separation of the components as shown in *Figure 28*. The values do not converge so we need to ensure DF is at the maximum but with separation of components. Our value was modeled after the shortcut design which gave us a precise value for DF.

Decreasing and Increasing Feed Stream Position (from 30)

Decreasing FP to 20:

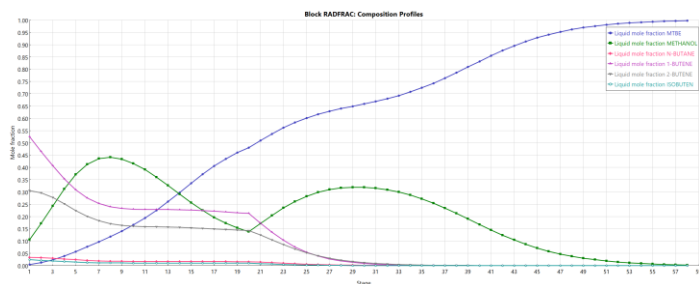


Figure 29 – Composition Profile and Product Results (FP 30)

Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
Units						MTBE2
Phase	Liquid					
Temperature	C					189.797
Pressure	bar					20
Molar Vapor Fraction	0					
Molar Liquid Fraction	1					
Mass Entropy	cal/gm-K					-1.48565
Molar Density	mol/cc					0.00538249
Mass Density	gm/cc					0.47371
Average MW						88.0094
♦ Mole Flows	kmol/hr					79.537
– Mole Fractions						
MTBE						0.9975
METHANOL						0.00249999
N-BUTANE						3.98287e-08
1-BUTENE						4.47333e-08
2-BUTENE						2.65533e-07
ISOBUTENE						1.37266e-09
♦ Mass Flows	kg/hr					7000

Increasing FP to 40:

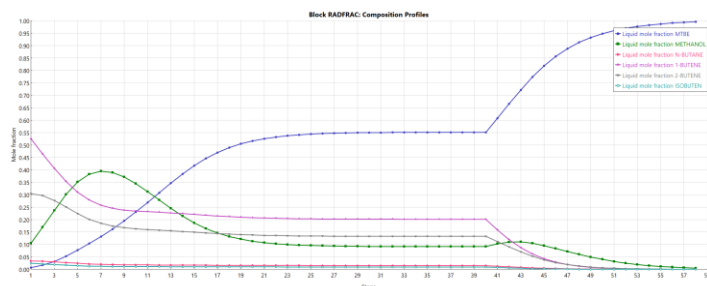


Figure 30 – Composition Profile and Product Results (FP 40)

Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
Units						MTBE2
Phase	Liquid					
Temperature	C					189.644
Pressure	bar					20
Molar Vapor Fraction	0					
Molar Liquid Fraction	1					
Mass Entropy	cal/gm-K					-1.48541
Molar Density	mol/cc					0.00539377
Mass Density	gm/cc					0.474098
Average MW						87.8973
♦ Mole Flows	kmol/hr					79.6385
– Mole Fractions						
MTBE						0.995352
METHANOL						0.00430319
N-BUTANE						2.6346e-05
1-BUTENE						0.000118018
2-BUTENE						0.000195414
ISOBUTENE						4.59978e-06
♦ Mass Flows	kg/hr					7000

If we decrease the feed stream position (FP) to 20 stages, we get purity of 99.8% MTBE and temperature of 189.8 °C as shown in Figure 29. Then if we increase FP to 40, this creates stages that are a waste of space with a purity of 99.5% and temperature 189.6 °C as shown in Figure 30. If we keep the amount of stages constant, then the FP position will seem to be ideal to be at the 20th stage but increasing it to 30th will give a higher purity. After a certain value of stages, it seems that increasing FP brings down the purity. Having a good balance of FP and total number of stages gives better purity.

Decreasing and Increasing Pressure (from 20 bar)

Decreasing P to 10 bar:

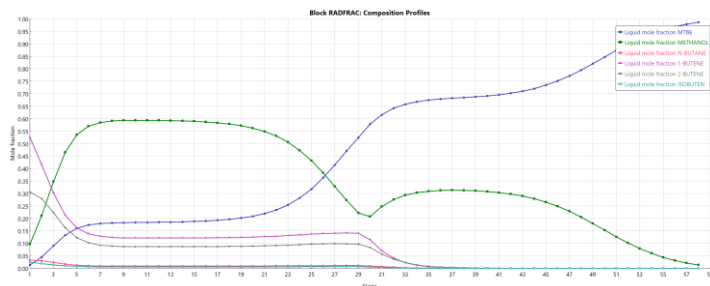


Figure 31 – Composition Profile and Product Results (P 10)

Increasing P to 30 bar:

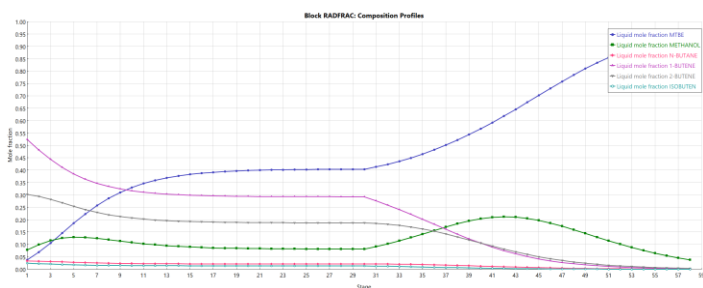


Figure 32 – Composition Profile and Product Results (P 30)

Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
						MTBE2
Phase						Liquid
Temperature						148.37
Pressure						10
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.55105
Molar Density						0.0063367
Mass Density						0.553721
Average MW						87.3831
Mass Flows						80.1112
Mole Fractions						
MTBE						0.986337
METHANOL						0.0136626
N-BUTANE						1.53605e-09
1-BUTENE						1.36567e-09
2-BUTENE						1.3987e-08
ISOBUTENE						4.09447e-11
Mass Flows						7000.36

Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
						MTBE2
Phase						Liquid
Temperature						214.38
Pressure						30
Molar Vapor Fraction						0
Molar Liquid Fraction						1
Mass Entropy						-1.43049
Molar Density						0.00456166
Mass Density						0.391604
Average MW						85.8468
Mass Flows						81.5407
Mole Fractions						
MTBE						0.956463
METHANOL						0.0377609
N-BUTANE						0.00044072
1-BUTENE						0.00195927
2-BUTENE						0.00330088
ISOBUTENE						7.52157e-05
Mass Flows						7000.01

Decreasing pressure to 10 bar gives a purity of 98.6% MTBE composition and temperature 148.4 °C as shown in Figure 31. Increasing the pressure to 30 bar gives a purity of 95.6% MTBE and temperature of 214.4 °C as shown in Figure 32. There is still good separation going on but adjusting the pressure to higher values results in a decrease in purity from 20 bar. But as we decrease the pressure, since the composition profile goes back, it seems as it will never reach a composition higher than 98.6%. Depending on customer specification, the correct pressure must be modeled to meet the purity of product.

References

- ^[1]Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design. Towler, Gavin and Sinnott, Ray. Elsevier, 2013. 2nd Edition.
- ^[2]<https://booksite.elsevier.com/9780080966595/exceltemplates.php> (reference for excel spreadsheet)
https://www.engineeringtoolbox.com/saturated-steam-properties-d_457.html (reference for steam properties)Transport Processes and Separation Principles. Geankopolis, Christie J., Hersel, A. Allen, Lepek, Daniel H. Prentice Hall, 5th Edition.
- ^[3]Eliceche, Ana M., et al. "Optimisation of Azeotropic Distillation Columns Combined with Pervaporation Membranes." Computers & Chemical Engineering, vol. 26, no. 4-5, 2002, pp. 563–573., doi:10.1016/s0098-1354(01)00775-x.
- ^[4]Ternary and Quaternary Liquid–Liquid Equilibria for Fuel Additives of the Water + Methanol + Toluene and Water + Methanol + Toluene + Methyl tert-Butyl Ether or tert-Amyl Methyl Ether Systems at 298.15 K. Kazuhiro Tamura*, Yao Chen, and, and Toshiro Yamada Journal of Chemical & Engineering Data 2001 46 (6), 1381-1386 DOI: 10.1021/je0100185
- ^[5]Thermodynamics of Octane-Enhancing Additives in Gasolines: Vapor–Liquid Equilibrium of the Ternary Mixtures Methyl tert-Butyl Ether + Heptane + Benzene and Methyl tert-Butyl Ether + Benzene + 1-Hexene at 313.15 K. José J. Segovia, María C. Martín, César R. Chamorro, and, and Miguel A. Villamañán*. Journal of Chemical & Engineering Data 1998 43 (6), 1014-1020. DOI: 10.1021/je980107q